Assignment 2: Q-Learning on a Tabular GridWorld

E1-277 Reinforcement Learning

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Problem Setup

The task is to train a Q-Learning agent on a deterministic 5×5 GridWorld environment. The state space is

$$\mathcal{S} = \{0, 1, \ldots, 24\}$$

where states encode grid coordinates in row-major order. The action space is

$$\mathcal{A} = \{\texttt{UP}, \texttt{DOWN}, \texttt{LEFT}, \texttt{RIGHT}\}.$$

Each action deterministically moves the agent to a neighbouring cell. Attempting to move outside the grid boundary or into a Wall cell leaves the agent's state unchanged (deterministic bouncing).

The environment contains five types of cells whose rewards are summarized in Table 1.

Symbol	Role	Reward	Terminal	Location
`S`	Start	-0.1	No	$(0, 0)$
`.`	Empty	-0.1	No	(various)
`G`	Goal	$+10$	Yes	$(4, 4)$
`P`	Pit	-5	Yes	$(3, 2)$
`W`	Wall	(stay)	No	(various)

Table 1: GridWorld cell types and rewards.

The agent is trained using the Q-Learning update rule

$$Q(s, a) \leftarrow Q(s, a) + \alpha\Big(r + \gamma \max_{a'} Q(s', a') - Q(s, a)\Big),$$

with parameters

$$\alpha = 0.1, \qquad \gamma = 0.99.$$

Exploration is performed using an ε-greedy policy with exponential decay

$$\varepsilon_t = \max(0.01, \varepsilon_0 \cdot 0.99^t)$$

over 500 episodes.

Remark 1 The small step penalty encourages the agent to reach the goal using the shortest path rather than wandering through the grid.

1 Part C — Convergence Analysis

1.1 C.1 Learning Curve

Figure 1 shows two traces: (i) the raw episode reward and (ii) a rolling average with window size $w = 50$.

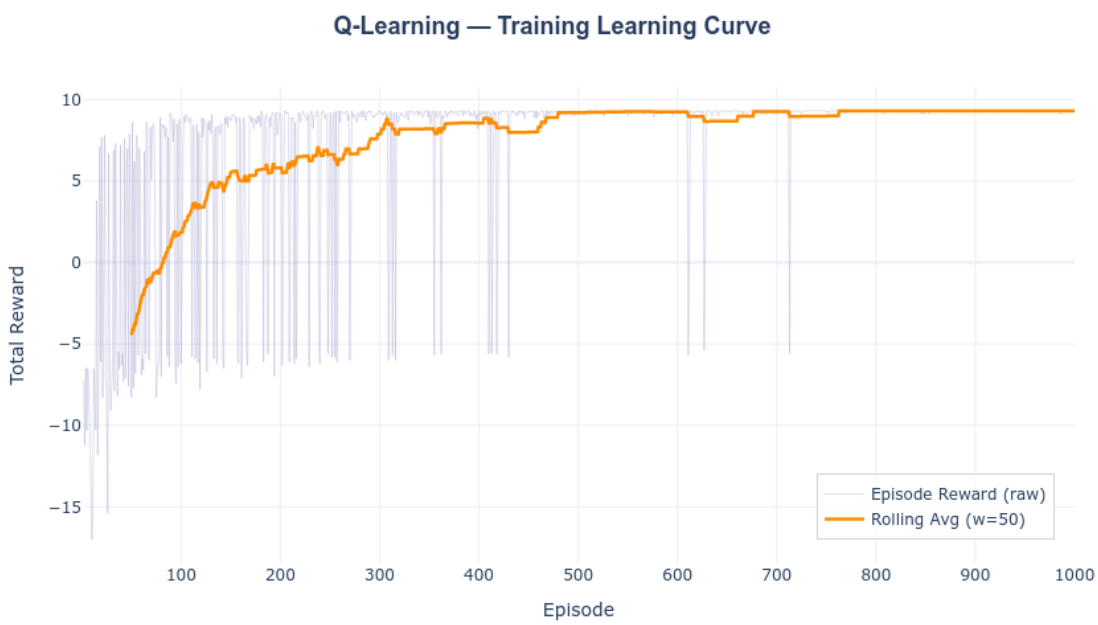

Figure 1: **Q-Learning training curve over 500 episodes.** The faint grey trace shows raw episode rewards. The orange line represents the 50-episode rolling average.

The rolling average eventually stabilizes near $+9.3$, indicating that the agent consistently follows the optimal trajectory.

1.2 C.2 When Did the Agent Converge?

The rolling-average reward increases rapidly during the first 200 episodes. After this point the curve begins to flatten and the agent consistently achieves high rewards.

Between episodes 300 and 350, the rolling average remains within

$$[9.1, 9.3].$$

The tail standard deviation of the final 50 rolling-average values is

$$0.1084,$$

which is well below the convergence threshold of 0.5.

Therefore the learning process can be considered to have converged around episodes **300–350**.

1.3 C.3 Why Is Early Training Noisy?

During the initial phase of training the exploration parameter satisfies

$$\varepsilon \approx 1,$$

which means the agent chooses actions almost uniformly at random.

This produces highly variable outcomes:

- The agent may reach the goal and obtain $+10$.
- It may fall into the pit and receive -5.
- It may wander through the grid accumulating step penalties.

As ε decays according to $\varepsilon_t = 0.99^t$, the agent increasingly exploits the learned Q-values. Once ε falls below approximately 0.1 (around episode 230), the agent almost always follows the greedy path to the goal, and the reward curve stabilizes.

2 Part D — Visualisations

2.1 D.1 Greedy Policy

D.1.1 Extracting the Policy

After training, the greedy policy is obtained from the Q-table as

$$\pi^*(s) = \arg\max_{a \in \mathcal{A}} Q(s, a), \quad \forall s \in \mathcal{S}.$$

For each state s, the action with the largest Q-value in row $Q[s]$ is selected.

D.1.2 Visualisation

Figure 2 visualizes the learned greedy policy. Each grid cell contains an arrow indicating the greedy action.

The optimal trajectory discovered by the agent is

$$(0,0) \to (0,1) \to (0,2) \to (1,2) \to (2,2) \to (2,3) \to (2,4) \to (3,4) \to (4,4).$$

This path requires 8 steps and yields total reward

$$10 - 7 \times 0.1 = 9.3.$$

The policy avoids the pit at $(3,2)$ and routes around the wall column.

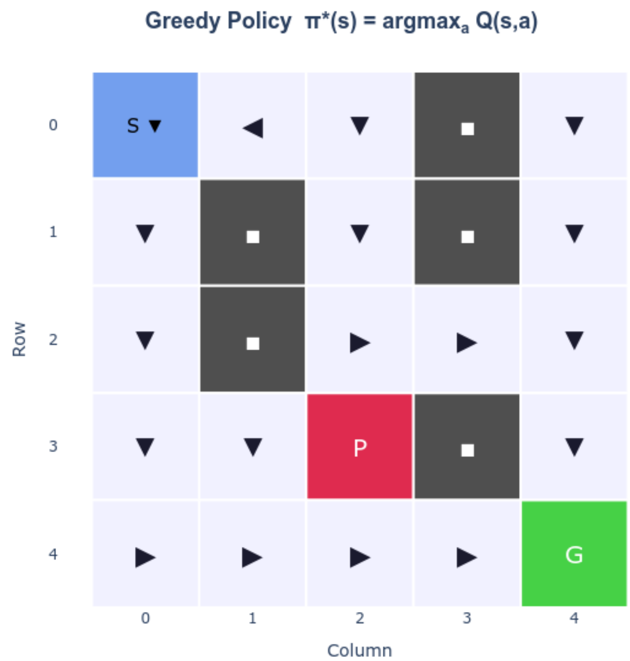

Figure 2: Greedy policy after 500 training episodes. Arrows indicate the optimal action in each non-wall state.

2.2 D.2 Value Function Heatmap

D.2.1 Computing the Value Function

The state-value function is computed from the learned Q-table using

$$V(s) = \max_{a \in \mathcal{A}} Q(s, a).$$

This quantity represents the expected cumulative discounted reward obtained when starting from state s and following the greedy policy.

D.2.2 Heatmap

Figure 3 shows the value function over the grid.

D.2.3 Interpretation

The heatmap reveals a clear value gradient originating from the goal:

- **Goal state $(4, 4)$:** Terminal state with value 0.
- **Neighbouring state $(3, 4)$:** Has value approximately 9.8, reflecting the large reward obtained after one step.
- **States along the optimal path:** Values decrease smoothly with distance from the goal, consistent with

$$V(s) \approx \gamma^d \cdot 10$$

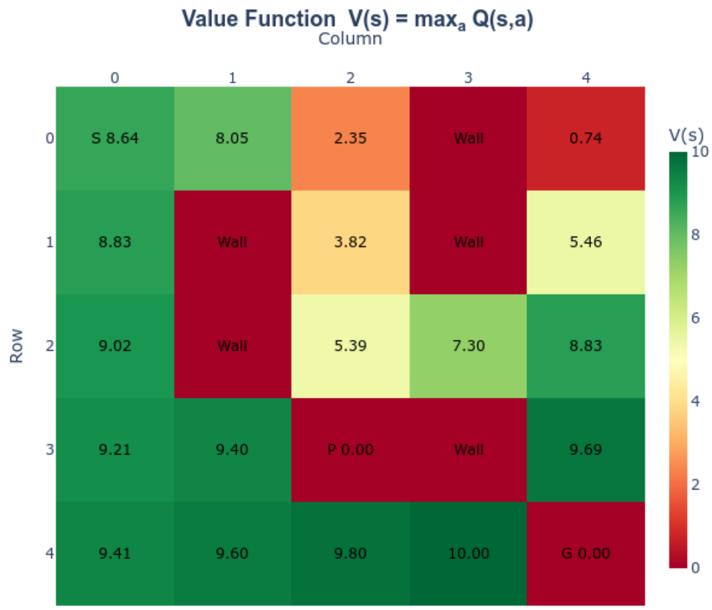

Figure 3: Value function heatmap derived from the learned Q-table.

where d is the number of steps to the goal.

- **Pit state** $(3, 2)$**:** Terminal state with negative reward -5.
- **Dead-end states (bottom-left corner):** These states have relatively low value because reaching the goal requires a long detour around the wall column.

Remark 2 The spatial structure of the value function confirms that the agent has learned a consistent representation of long-term return in the environment.

Notation

Symbol	Description
GridWorld Environment	
S	Start state located at $(0, 0)$
G	Goal state located at $(4, 4)$
P	Pit state located at $(3, 2)$
W	Wall cell blocking movement
$\mathcal{S}$	Set of all states in the grid world
$\mathcal{A}$	Action space $\{\texttt{UP}, \texttt{DOWN}, \texttt{LEFT}, \texttt{RIGHT}\}$
Q-Learning Variables	
$Q(s, a)$	Action-value estimate for state-action pair
$\pi(s)$	Policy mapping states to actions
$\pi^*(s)$	Greedy policy $\arg\max_a Q(s, a)$
$V(s)$	State-value function $\max_a Q(s, a)$
Learning Parameters	
α	Learning rate for Q-value updates
γ	Discount factor for future rewards
ε	Exploration probability in ε-greedy policy
ε_t	Exploration rate at episode t
w	Rolling-average window size for reward smoothing
Training Metrics	
R_t	Total reward obtained in episode t
$\bar{R}$	Rolling average episode reward
d	Number of steps from state s to the goal

Table 2: Notation reference for symbols used in the Q-Learning GridWorld experiments.